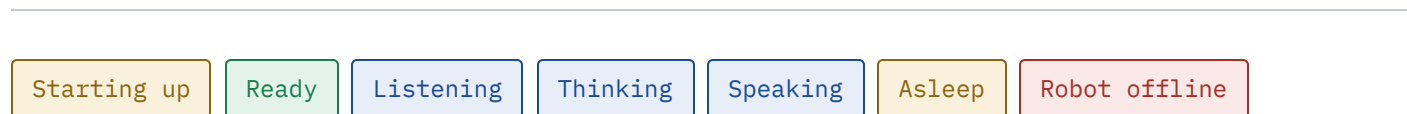


Reachy Field Handbook

Everything needed to set the robot up, run it in front of visitors, and understand every button on its dashboard. Written for whoever is standing next to it — no technical knowledge assumed.



What you have

Two pieces of equipment that talk to each other over WiFi, and they must be on the **same network** to work at all.

PIECE	WHAT IT DOES
The robot Reachy Mini	The ears, mouth, eyes and body. Its microphone hears the room, its speaker talks, its camera sees faces, and six motors move its head and antennas. It does very little thinking of its own.
The laptop	The brain. Speech recognition, the voice, the AI and the dashboard all run here. This is why replies come back in a few seconds rather than twelve — the robot's own small computer would be far slower.

Everything you control lives on one web page — **the dashboard** — which you open on the laptop or on your phone.

DOES IT NEED THE INTERNET?

Mostly no. Hearing, speaking, face recognition and every scripted answer run entirely on the laptop. Only the *AI* answers normally go to the cloud, and if the internet drops the robot automatically falls back to a smaller AI running locally — slower and less sharp, but still working. The one feature that genuinely stops without internet is [Look at this](#), which needs to send a photograph.

First-time setup

Only needed once per laptop. If somebody has already set this machine up, skip to [Starting it](#) — you will know because there is a `Start Reachy` icon on the desktop.

1 Put the robot on the same WiFi as the laptop.

This is the single most common cause of "it won't start". At Trinity that usually means both on `TCD_IoT`. The robot cannot be found across two different networks, and it cannot be found on the guest network at all.

2 Run the installer.

Right-click `install.ps1` in the project folder and choose *Run with PowerShell*. It installs everything the robot needs, downloads the speech models, and prepares the wake phrases. It prints a step as it finishes each one, and stops with a red message if anything fails.

3 Add the AI key.

The installer creates a file called `.env`. Open it in Notepad and put the Anthropic API key after `ANTHROPIC_API_KEY=`. Without it the robot still works, but only with the smaller offline AI. This file is private and is never uploaded anywhere. If you want to lock the dashboard, this is also where `DASHBOARD_PASSCODE=` goes — see [Every control](#).

4 You are done.

The installer creates a `Start Reachy` icon and a `Reachy Dashboard` icon on the desktop, and sets the robot to start by itself whenever this user signs in — so a visitor meets a working robot rather than a desktop icon.

IF YOU EVER WANT TO STOP IT AUTO-STARTING

Delete `Reachy Autostart` from the Windows Startup folder. Everything else keeps working; you just launch it yourself with the desktop icon.

Starting it

1 Turn the laptop on and sign in.

Reachy starts by itself a few seconds later.

2 Power on the robot.

Give it about a minute to join the WiFi.

3

Wait for the wiggle.

When the robot looks left, right and nods, it is ready. That takes a minute or two from sign-in — mostly loading the speech software.

4

Open the dashboard.

Double-click `Reachy Dashboard` on the desktop, or type `localhost:8080` into the browser.

If it did not start by itself, or you closed the black window, double-click `Start Reachy`. You never have to type the robot's address — it goes and finds it, and prints where it found it.

What the black window is telling you

MESSAGE	MEANING	WHAT TO DO
<code>Looking for the robot ... found at 10.19.x.x</code>	Normal. It located the robot and is connecting.	Nothing.
<code>NOT FOUND</code>	No robot answered anywhere on this network.	Check the robot is powered on, and that it and the laptop are on the <i>same</i> WiFi. Then <code>Start Reachy</code> again.
<code>UNREACHABLE</code>	It answered a moment ago and has now dropped off.	Restart the robot, then <code>Start Reachy</code> again.
<code>Media acquired from daemon</code>	Normal. It took control of the robot's microphone and camera back from any previous session.	Nothing.
<code>Ready. Say a wake phrase to talk.</code>	Fully up.	Say <code>Hey Reachy</code> .

Stopping and watching it

Close the black window to stop the robot, or press `Ctrl+C` in it. To watch what it is hearing and saying as a live feed, run `watch_reachy.ps1` — it is read-only and safe to start and stop at any time, including while a visit is running.

Using it from your phone

Put the phone on the same WiFi as the laptop, then type the address the black window printed when it started — something like `http://10.19.4.73:8080`. Handy when you are standing with visitors and the

laptop is across the room. Two people can have it open at once.

Talking to it

Say `Hey Reachy`, wait for the antennas to perk up, then ask. **One wake phrase per question** unless you turn on `Open mic`. This is the thing visitors get wrong most often — they say it once and keep talking.

It answers to its name, however you say it

All of these wake it:

`Hey Reachy` `Hi Reachy` `Hello Reachy` `OK Reachy` `Hello there Reachy` `Attention Reachy` `Reachy`

It also answers to `Ricky`, `Richie` and `Ritchie` on purpose — plenty of Irish and Scottish speakers land nearer those, and nobody should have to say the name carefully to be heard. The cost: if somebody in the room is actually called Ricky or Richie, the robot occasionally thinks it was called, listens, hears nothing meant for it, and goes back to waiting.

Interrupting and stopping

TO DO THIS	SAY	WHAT HAPPENS
Cut it off mid-answer	<code>Hey Reachy</code> while it is still talking	It stops within about a second and listens. Your question is already captured — you do not need to repeat it.
Send it quiet	<code>go to sleep</code> · <code>goodbye</code> · <code>turn off</code> · <code>stop listening</code> · <code>power down</code> · <code>shut down</code>	It says goodnight and stops responding — but keeps listening for its name, so any <code>Hey Reachy</code> brings it back.
Fix a name it got wrong	<code>that's not my name, it's Sarah</code> · <code>call me Sarah</code>	It corrects itself immediately, in any mode.

When it asks you a question

Whenever Reachy ends an answer by asking you something, it listens for your reply **without needing the wake phrase** for about half a minute. Take your time — a thought-out answer still counts. A one-word reply like `yes` needs to come in the first ten seconds or so; after that, answer in a short sentence.

Reading the screen

Under the title is one wide bar saying what the robot is doing right now. It changes colour with the state so you can read it across a room.

BAR SAYS	MEANING
Starting up	Loading. A minute or two. Normal.
Ready	Waiting for someone to say its name.
Listening	It heard its name and is taking your question.
Thinking	Working out the answer.
Speaking	Talking. You can interrupt.
Asleep	Somebody said goodbye. Normal, not a fault — say its name.
Robot offline	The laptop lost the robot, usually WiFi. See When it misbehaves .

Below the bar is the **mode grid**. The demonstration that is running is the one **solid coloured block** on the page and nothing else is ever solid — one glance, one colour, one answer.

Everything the robot hears and says appears in the transcript as it happens, including the words appearing one at a time as you speak. That live text is the fastest way to tell a microphone problem (nothing appears) from a mishearing (the wrong words appear).

Every control on the dashboard

In the order they appear down the page. Type a word to find the one you are looking at.

The top bar

Always visible, above everything

Status line

Read-only. The big word is what the robot is doing; the small line underneath explains it. Deliberately silent to screen readers — it would chatter all visit.

● Recording

A red blinking pill that **only appears while a research session is actually recording**. Press it to stop recording.

Stopping keeps the data — deleting is a separate deliberate act in the Research panel. It checks every five seconds regardless of which panels are open, because it exists for the room, not for you.



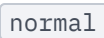
volume

0–100 in steps of five. Sets the robot's speaker.

Applies when you let go of the slider, not while dragging. Shows  if the robot is unreachable. Kept out on the surface because it is the one thing you change mid-visit.



speaking speed

How fast it talks, shown in words:  /  /  / .

Takes effect on the very next line. Saved on the robot, so it survives a restart or a WiFi drop.

⚙ Settings

Opens the settings tray.

Only exists on phones — on any screen wider than about 760 pixels the tray is permanently open and this button is hidden.

up 12m 30s

Read-only. How long the robot has been running.

Settings

Behind 🌀 on a phone, always open on a laptop

model

Read-only. Which AI is answering and whether it is **in the cloud** or **on this laptop**.

Check this before a visit. "On this laptop" means the internet or the key is down and it has fallen back to the smaller offline AI.

Web search

Off by default. Lets it look things up online — weather, news, anything current. Slower, and costs a little each time.

Greyed out when the AI is running locally, because a local model cannot search. Makes a good demonstration in itself: ask for today's weather off, then on.

Open mic

Off by default. Say the wake phrase once to start, then keep asking — no wake phrase per question, until about half a minute of quiet.

Better one-to-one. Leave it *off* for a crowd: with several people talking, the wake phrase is what keeps the robot out of other people's conversations.

personality

Default, Professional, Friendly, Consultant.

This changes the robot *everywhere* — its wording, its speaking voice and how it holds its head — not just in the AI Personality demo.

voice

Which of the installed voices it speaks in.

Empty for the first minute after starting, while the models load — it fills itself in. A personality brings its own voice and overrides a hand-picked one.

Appearance

Auto / Light / Dark.

This device only. Changes nothing on the robot.

Colour

Five themes: Trinity (the default), Violet, Teal, Rose and **High contrast**.

This device only. High contrast is the one for a sunlit atrium.

Clear answers

Makes it forget answers it has memorised, so the next question goes to the AI fresh.
Only needed if it keeps repeating something wrong. No confirmation — but it is harmless.

Mode

The grid of demonstration buttons

A mode button

Switches the robot to that demonstration, instantly — you do not have to wait for it to finish talking.

It also wakes the robot if it was asleep. Exactly one button is filled solid and reads `Running`; nothing else on the page is ever solid.

A greyed-out button

Says why underneath — usually `needs faces`, `needs study`, or that it failed three times in a row and was set aside.

Put back

Appears only on a button that was set aside after failing. Re-enables it.
Saves you restarting mid-visit.

Arrange

Turns on folder editing: drag handles, a `Put ... in` dropdown on every button, and `+ New folder`.

While arranging, pressing a button does *not* start it — a thumb that slides must never launch a demo in front of fifteen students. It lets go by itself after 90 seconds. Hold one button on top of another for half a second to group them, phone-home-screen style. Deleting a folder never deletes a button.

Folders

Tap the arrow to collapse one. The demo that is *running* stays visible even inside a collapsed folder.

The arrangement lives on the robot, so it survives a restart and everyone's dashboard shows the same one. Twelve folders maximum. There is no undo.

Direct control

Under the mode grid

Type box + **Say it**

Makes the robot say whatever you type, up to 300 characters. Enter also sends.

Wakes it if asleep. The text stays in the box if it failed, so you can retry. Invaluable when the room is too loud for the microphone.

 **Listen now**

Starts listening exactly as if somebody had said the wake phrase.

The wake word is the weakest link in a loud room; this is the way round it.

Say any of these

A read-only reference panel

Phrase chips

Every phrase the robot actually matches: how to wake it, the trigger phrases for each demonstration, and how to send it to sleep.

Read straight off the robot, so it can never disagree with what really works. Worth showing visitors.

Sounds

Collapsed panel — click the heading to open

Six buttons

applause **correct** **fanfare** **ready** **uhoh** **wrong** — press one to play it on the robot.

Generated by the robot, not recordings, so there is nothing to license. They queue with the voice, so a sound plays between turns rather than the instant you press.

Is it well?

Collapsed panel. Says `all well` or `2 things to look at` in the heading

The health table

Read-only, refreshing every ten seconds while open: the robot's state, its control-loop speed, motor errors, its address, firmware and serial number.

Battery

Always reads *not reported by this robot* — the hardware exposes no charge level.

Not a fault. Keep an eye on the cable instead.

What it cannot tell you

The speaker, microphone, camera and movement can only be tested by running

`Check it works` in the mode grid.

How the visit went

Collapsed panel

Three tiles

`Things said to it`, `Demos used`, `Different questions` — today's totals.

`Which demos ran` • `Most asked`

Bar lists of what actually happened.

Questions it could not answer

Real questions where the robot said it did not know — the exact list worth teaching it.

Aggregate only: the question, never who asked it. The [Teach Reachy](#) panel puts a `Teach` button beside each one.

What it hears & says

The transcript, at the bottom

Download

Saves the whole session's conversation as a text file.

Logs

Off, you see a readable conversation. On, it also shows wake-word hits, partial transcriptions and status messages.

Remembered per device. Turn it on when diagnosing; off in front of visitors.

The transcript

Everything heard and said, timestamped. It only follows along automatically if you were already at the bottom — scroll up to read and it will not yank you back.

This dashboard is locked

Only appears if a passcode is set and this browser has not entered it

Passcode + **Unlock**

Unlocks this browser for twelve hours.

Set it by putting `DASHBOARD_PASSCODE=` in the `.env` file and restarting. No passcode set means no lock, which is the default.

IF THE DASHBOARD LOOKS BROKEN, SCROLL DOWN

A locked dashboard shows an empty mode grid and dead panels, because every panel was refused. The unlock box sits *below* the grid. Also worth knowing: **restarting the robot logs everybody out**, and the passcode is one shared code on plain WiFi — a lock on a door, not a safe. The robot keeps talking to visitors either way.

What is shared, and what is just yours

SETTING

Appearance, Colour, Logs

LIVES

In this browser

SO...

Your phone and the laptop can look different. Changing them affects nobody else.

SETTING	LIVES	SO...
Folders and their names	On the robot	Everyone sees one grid. Survives a restart.
Speaking speed	On the robot	Survives a restart and WiFi drops.
Web search, Open mic, personality, current mode	On the robot, in memory	Shared across dashboards, but reset by a restart .
Features, taught answers, research data, visit stats	The robot's database	Backed up daily, seven days kept.

Nothing on this page ever needs the robot restarted. New buttons appear immediately, taught answers take effect immediately, and Put back exists so a failed demo does not cost you a restart mid-visit.

Every mode

Press one on the screen, or say its phrase out loud. Switching is instant. The robot starts in **AI Conversation** and falls back to it whenever anything else stops working.

The three to show first

AI Conversation no phrase — it is where everything lands

The normal one. Any question nothing else claims is answered here, with the Hub's real facts in front of the AI so it does not invent staff or projects in a room containing them. It answers the time and date itself, instantly, without asking the AI.

Try: `who runs the AI XR Hub?` then `what is extended reality?`

AI Personality say `switch personality`

Answers the *same* question again as a different character — that contrast is the whole demonstration. You hear the difference three ways at once: the wording, the speaking voice, and how it holds its head. Hub facts are deliberately withheld here, so a Hub question gets "ask your host" rather than a confident invention.

Ask something arguable — `should a business student learn to code?` — then change the  dropdown twice. Factual questions make all three sound the same; arguable ones make the difference obvious.

Business Brainstorming say `let's brainstorm`

Built for four people crowded round the robot, not one person at a desk. It writes each question from what the group just said rather than working down a list, then offers three concrete directions back. It holds the microphone open while waiting, so nobody says the wake phrase to answer.

Let them drive it. To skip ahead, anyone can say `what do you think?` ; for a wrap-up, say `summarise that` .

For a group arriving

Welcome say `do the welcome`

About thirty seconds explaining the Hub, one line at a time so it stays interruptible. It waits up to twenty seconds for a face before starting anyway, and will not repeat itself for the same audience.

Dean's Office say `do the deans office`

A two-hander for a live event: the Dean asks the questions, Reachy waits in complete silence and answers each one from the approved script. It holds the microphone the whole time, so nobody says a wake phrase, and it ignores everything that is not a cue rather than chiming into the Dean's lines.

From beside the stage: `next` plays the next block if a cue is missed, `say it again` repeats the last one, `start again` rewinds silently.

About Reachy say `tell me about yourself`

Explains the robot to non-engineers, honestly, and answers the four questions visitors always interrupt with — how it hears you, whether it can see you, which AI that is, and whether it needs the internet. Those answers are read from what is actually running this morning, not from a fixed script.

Quiz the group say `quiz us`

Four questions about the Hub and AI, keeping score. Answers are matched loosely on purpose — thirty teenagers shouting at once produces messy transcripts, and a quiz nobody can win is no quiz. Shuffled, so the second group of the day gets a different order. Wrong answers still get told the fact.

One-to-one

Which masters fits me? say `which masters`

Three short questions, then it names one or two of the thirteen taught masters and why each. It knows the real programmes in detail, and it will **refuse** fees, deadlines, entry requirements and whether somebody would get in — those change yearly and belong to you, not to a robot. Every path ends by pointing at a human.

Vision & Face Tracking say `show me face tracking` · needs the camera

Head-following runs under every mode all the time; what this adds is putting a name to a face. After somebody has stood still a few seconds it offers, once, to learn their name, and holds the microphone open for the whole exchange. Nothing is stored until you say yes to the name it reads back. It keeps learning your face over time, so it recognises you more reliably the more often you are around.

Say `that's not my name, it's Sarah` any time, in any mode, to correct it.

Look at this say `what is this?` · needs the camera and the internet

Hold an object up and ask. It says "let me take a look" first, takes one picture, and names what it sees in a sentence. This is the only feature that sends an image anywhere — see [Privacy](#). With no internet it says so plainly rather than guessing.

Atmosphere and utilities

Storyteller say `tell me a story`

A short made-up story, told in a storyteller's voice, on entering, on request, and every few minutes. Asked for a story *about the Hub* it declines and answers normally instead, because the storyteller has no Hub facts and would cheerfully invent staff and funding.

Greet in another language say `say hello in Spanish`

Hello in Spanish, French, German, Italian or Irish: a greeting, one true sentence, then an explicit hand-back to English. Greeting only, on purpose — it cannot *understand* those languages and must never appear to.

Dance say `let's dance`

Loops a head-and-antenna routine with the occasional dry aside — "All neck. No legs." Stop it by saying `stop`, `enough` or `settle down`, or by switching mode.

Timers say `set a timer for five minutes`

Spoken timers from natural phrasing. They keep counting and **still ring after you switch to another demonstration, and even while the robot is asleep.**

Idle click it — no phrase

Sits quietly and starts nothing by itself. Use it during a talk, in a photo, or when the robot is being explained rather than used. It is not asleep — it still answers if spoken to, and another mode's phrase still switches to that mode, so a speaker can hand the room back to the robot by voice.

Before and after a visit

Check it works say `run the check`

Half a minute of self-test: speaker, movement, camera, microphone, the AI, then a verdict, ticking off a list on the dashboard as it goes. It asks you to say something so it can check its hearing. It reports the signal level it saw rather than claiming it understood you — a check that fails in a quiet room is one staff learn to ignore.

Research session greyed out until Research mode is armed

Records a conversation for HRI research. Deliberately unreachable unless somebody has armed recording on the dashboard, so it cannot be stumbled into on an open day. See [Research mode](#).

YOUR OWN BUTTONS APPEAR HERE TOO

Anything built in `Build a feature` shows up in the same grid alongside these — that is how the `Erasmus welcome` button got there.

The Hub answers

Five questions about the Hub have **fixed, approved answers** written with the Dean. Whoever asks, in whatever mode, the robot speaks these word for word and instantly — there is no AI involved and no waiting.

ASK ANY OF THESE

ANSWER BEGINS

`would you like to introduce yourself` · `who are you`

"Of course! Hello new friends, and welcome to Trinity Business School..."

`what is the AI XR Hub` · `tell me about the hub`

"Well, you might expect a robot to talk about technology, but that's not really what we're about..."

`why are the students so important` · `why do human skills matter`

"Because as AI gets better, human skills matter more, not less..."

`what will students actually do in the hub`

"You'll get hands-on experience with immersive technology..."

`any final advice for our new students`

"Yes. Be curious. Try something unfamiliar..."

WHY SOME QUESTIONS STILL GET AN AI ANSWER

Questions starting **who, where, when, which** or **how** deliberately go to the AI instead — "who runs the Hub?" has a real specific answer (Professor Na Fu directs it, Professor Laura Berry manages it), and reciting the mission speech at it would read as a robot that was not listening. A long multi-part question also goes to the AI, so it can answer all the parts.

The order of precedence, which never changes: **the Hub script beats taught answers, and taught answers beat the AI improvising.**

Teaching it new answers

This is how the robot gets better over time, and it takes about thirty seconds. Open **Teach Reachy**.

1 Look at the bottom of the panel.

It could not answer these today lists real questions visitors asked where the robot said it did not know.

2 Press **Teach** beside one.

The editor opens with the question already filled in.

3 Type exactly what Reachy should say

, and optionally a few other ways people might ask the same thing, separated by commas.

4 Press **Save**.

From that moment the robot speaks your answer instantly whenever the question comes up — in any mode, with no restart.

A taught answer works two ways. Asked directly, it is spoken **word for word**. Asked as a paraphrase, the AI is shown your answer before it replies — so teaching it about parking once also improves "where do I leave my car".

Letting the AI draft them for you

Draft suggestions from unanswered questions asks the AI to write first drafts for today's list. They arrive **switched off**, marked **AI draft**, and are never spoken until you press **Approve**.

PLACEHOLDERS MUST BE FILLED IN FIRST

Anything the AI did not know it writes as `[bracketed placeholder]` — and `Approve` will refuse the answer until you edit them out. That is deliberate: the robot must never invent a fact about the Hub, and that includes a fact an AI drafted confidently.

What the panel refuses, and why

IT WILL NOT ACCEPT	BECAUSE
An answer over about 700 characters	That is more than half a minute of talking. A room stops listening.
A one-word phrasing	It would fire on fragments of unrelated conversation. A bare "welcome" once launched the whole welcome speech every time somebody said "you're welcome".
A phrasing containing another mode's trigger	Phrases like "what do you see" belong to the camera demo and are matched first, so the robot would switch modes instead of answering. It names the clash so you can reword.

Other buttons on each taught answer

BUTTON	WHAT IT DOES
<code>Edit</code>	Change the question, the answer, or the phrasings.
<code>Turn off</code>	Keeps it but stops the robot using it.
<code>Approve</code>	Turns a draft on, after re-checking it.
<code>Delete</code>	Forgets it for good.

A green dot means live. A hollow dot means saved but never spoken.

Building your own button

Open `Build a feature` to make a new button — most often a welcome for a particular group. Describe what you want in plain English and the assistant drafts the words; you edit them and press `Save`. **The button appears straight away and the robot does not need restarting.**

A feature can do four things, and only these four, on purpose:

STEP	WHAT IT DOES
SAY	Speaks a line, with an expression.
ASK	Asks a question, listens, and can let the AI answer the reply.
DANCE	A short dance.
WAIT	Pauses until somebody is standing in front of it.

A feature can also finish by handing over to another demonstration, so one button can welcome a group and then hand to the tour or the quiz. That has to be the last step.

`Disable` takes a button off the dashboard but keeps the script. `Delete` removes it for good.

TWO HONEST NOTES

Features are **not reviewed** — anyone who can open the dashboard can add one and the robot will say it to whoever is standing there, so put your name in the box. And the assistant writes `[bracketed placeholders]` for anything it does not know; fill those in before saving or the robot will read them aloud.

Research mode

For Professor Berry's human–robot interaction work. Put your name in `Armed by`, tick `Recording armed`, then run the `Research session` demo.

THIS IS NOT ETHICS APPROVAL

Trinity requires approval before collecting data from participants. This only makes the robot able to do it properly once you have that.

Switching it on is the consent. The robot does not read a notice or ask anybody out loud; it records as soon as the session runs. So take consent from participants *before* you switch it on, the way an approved protocol does it — on paper, before they sit down. The `Armed by` name is the only record of somebody accountable for the data, which is why it is required and written against every turn.

Anybody in the room can stop it by saying so — `stop recording` keeps what was said and stops; `delete my data`, `I changed my mind` or `I do not consent` **delete the session outright**. The sentence that stopped it is not recorded either.

It is **off after every restart**, on purpose. Since the robot restarts whenever the WiFi drops, check it is still armed if you have been running a long session — the red `REC` pill at the top of the dashboard is showing

whenever it is recording, and clicking that pill stops the recording.

Getting the recordings out

THEY ARE TRANSCRIPTS, NOT AUDIO

Nothing the robot hears is ever saved as sound — speech becomes text and the audio is discarded on the spot. There is no play button and no file to recover. That is the point: a transcript with no name and no voice attached cannot be traced back to the person who said it, even by whoever holds the database.

The `Recordings` list shows every session: when it started, the condition, who armed it, how many turns and the average time to first word. The live one is marked `live`.

BUTTON	WHAT IT DOES
<code>View</code>	Shows the transcript in the page, to check a session is what you think before exporting.
<code>CSV</code>	Downloads it for analysis — one row per turn, with persona, both timings, and which AI answered.
<code>Text</code>	The same thing laid out to read.
<code>Delete</code>	Removes that session outright. Use if a participant asks afterwards.

Two timings, and they mean different things. `first_word_s` is how long the participant waited before hearing anything — that is responsiveness. `latency_s` is the whole turn including the robot speaking, so it grows with how long the reply was. For a study of responsiveness use the first; the second makes a talkative robot look like a slow one. `backend` records whether the cloud or local AI answered, because a WiFi drop mid-session changes your condition without anybody choosing to.

Privacy, and what to tell visitors

What stays on the laptop

Face detection, face recognition and the list of who it knows **never leave the machine**. Speech is turned into text on the laptop too. The AI answers go to the cloud as text, with no name attached.

The one exception

`Look at this` is the only feature that sends a picture anywhere. When someone asks it to look at an object, it sends a single photograph to the same AI that answers its questions. Four things bound it:

- **Only when asked.** Nothing takes a picture on its own.
- **It says so first.** "Let me take a look" is spoken *before* the picture is taken.
- **Nothing is kept.** Sent, answered, dropped — never written to disk, never near the research data.
- **It is deliberately a poor photograph** — shrunk to 640 pixels, plenty to name a mug, not much use for recognising the people behind it.

If you are running anything where visitors have been told the robot does not photograph them, switch that mode off in the grid and it cannot be reached.

Honest limits worth saying out loud

- **It cannot move around** — no wheels, legs or arms. Head and antennas only.
- **It cannot play music** — only its own short sound effects.
- **It does not know today's news or weather** unless is switched on.
- **It will not guess about the Hub.** Asked something it does not know — which headsets are in the room, fees, entry requirements — it says so and points at a human. A robot inventing a plausible answer in front of the people who would know is worse than one admitting it does not know.

When it misbehaves

Nine times out of ten the fix is: close the black window, double-click `Start Reachy`.

WHAT YOU SEE	WHAT TO DO
It ignores <code>Hey Reachy</code>	Look at the screen — do your words appear as text? If yes the microphone is fine; try just <code>Reachy</code> on its own and pause before the question. If nothing appears at all, the robot has probably dropped off the WiFi.
<code>NOT FOUND</code> when starting	Laptop and robot are on different networks. Put both on the same WiFi and start again.
<code>UNREACHABLE</code>	The robot dropped off. Restart the robot, then start again.
It talks over itself, two voices	Two copies are running. Close every black window, then <code>Start Reachy</code> once.
It answers something nobody asked	<code>Open mic</code> is on and it picked up the room. Untick it — in a crowd the wake phrase is what keeps the robot out of other people's conversations.
It wakes when nobody said its name	Somebody nearby is called Ricky, Richie or Rachel. Deliberate, so visitors with accents are heard. It gives up after a moment.
Answers are slow or vague	Check the AI label in Settings. If it says the model is on this laptop, the internet or the API key is down and it has fallen back to the smaller offline AI. It still works, just less well.
A mode button is greyed out	It says why underneath. Usually a demo that failed a few times and was set aside — click it to re-enable, or restart.
It says <code>Sorry – I did not catch that</code> a lot	The room is loud. Ask people to stand closer, or use the mode grid rather than voice to switch demonstrations.
Frozen but still talking	Wait a minute — it restarts itself.
Nothing works after three minutes	Power the robot off and on, close the window, <code>Start Reachy</code> again. Still stuck? Restart the laptop. Truly stuck? Call whoever set it up.

Two-minute check before a visit

1 Sign in, robot on, wait for the wiggle.

2

Say **Hey Reachy, what is the AI XR Hub?**

It should start "Well, you might expect a robot to talk about technology..." almost immediately. That one test confirms the microphone, the voice, and the Hub script all work.

3

Check the AI label in Settings

says a model in the cloud, not on this laptop.

4

Set the volume for the room.

Too loud distorts; about 80% is usually right.

5

Leave it in **AI Conversation**

for visitors arriving, and untick **Open mic** if it is going to be a crowd.

Trinity AI XR Hub, Trinity Business School. Keep this open on a phone during a visit — the contents list on the left jumps straight to any section.